

Project Overview

This project performs end-to-end data quality assessment, data cleaning, and customer segmentation using RFM (Recency, Frequency, Monetary) analysis on a retail bicycle sales dataset. The goal is to transform raw, dirty data into clean, analysis-ready datasets and derive actionable customer insights for business decision-making.

Dataset Description

The project works with four primary datasets extracted from Raw_data.xlsx:

Dataset	Records	Features	Description
Customer Address	3,999	6	Customer location data (address, postcode, state, country, property valuation)
Customer Demographics	4,000	13	Customer personal info (name, gender, DOB, job, wealth segment, tenure, etc.)
New Customer List	1,000	18	Prospective/new customer data with additional ranking metrics
Transactions	20,000	13	Sales transaction records (product, brand, pricing, order details)

Notebook 1: DQA and Data Cleaning — Customer Address

Purpose: Performs Data Quality Assessment (DQA) and cleaning on the CustomerAddress dataset to ensure accurate geographic and property valuation data.

Steps Performed

1. Data Loading & Initial Inspection

Loaded data from Excel (Raw_data.xlsx, sheet: CustomerAddress). Verified data types: customer_id (int64), address (object), postcode (int64), state (object), country (object), property_valuation (int64).

2. Missing Values Check

No missing values found across all 6 columns.

3. Inconsistency Detection & Correction

The state column had inconsistent naming — "New South Wales" (86 records) and "NSW" (2,054 records); "Victoria" (82 records) and "VIC" (939 records). Standardized all state names to abbreviations (NSW, VIC, QLD). Result: 3 unique states — NSW (2,140), VIC (1,021), QLD (838).

4. Duplication Check

Checked for duplicate records excluding the primary key (customer_id). No duplicates found (3,999 unique records).

5. Output

Exported cleaned dataset to CustomerAddress_Cleaned.csv.

Key Insight: NSW has the highest customer concentration (~53.5%), followed by VIC (~25.5%) and QLD (~21%). This indicates a strong presence in the New South Wales market.

Notebook 2: DQA and Data Cleaning — Customer Demographics

Purpose: Cleans and prepares the CustomerDemographic dataset by handling missing values, removing irrelevant columns, standardizing data, and deriving new features.

Steps Performed

1. Data Loading & Initial Inspection

Loaded 4,000 records with 13 columns. Identified data types and initial quality issues.

2. Dropping Irrelevant Columns

Removed default column — contained irrelevant/malformed data (e.g., HTML scripts, shellshock code, timestamps).

3. Missing Values Analysis

Column	Missing Count	Missing %	Action Taken
last_name	125	3.1%	Filled with "None"
DOB	87	2.2%	Dropped records (low %)
job_title	506	12.7%	Retained (acceptable)
job_industry_category	656	16.4%	Retained (acceptable)
tenure	87	2.2%	Dropped along with DOB records

4. Gender Standardization

Issue: Inconsistent gender labels ("Male", "F", "Female", "M", "U"). Fix: Standardized to Male, Female, U (Unknown).

5. Feature Engineering – Age Calculation

Created Age column from DOB using: $\text{Age} = \text{Today.year} - \text{DOB.year} - \text{adjustment}$. Converted DOB to proper datetime format.

6. Job Title & Industry Category Cleaning

Replaced missing job_title values with "Missing". Replaced missing job_industry_category values with "Missing".

7. Output

Final cleaned dataset: 3,912 records × 13 columns. Exported to CustomerDemographic_Cleaned.csv.

Key Insights

- Age Distribution: Customers span a wide age range, with concentrations in middle-age brackets.
- Gender Distribution: Mix of Male, Female, and Unknown gender customers.
- Wealth Segments: Three segments identified — Mass Customer, Affluent Customer, High Net Worth.
- Data Quality Issue: The default column contained malicious/test data (XSS scripts, shellshock payloads), indicating potential data entry/validation issues in the source system.

Notebook 3: DQA and Data Cleaning — New Customer List

Purpose: Cleans the NewCustomerList dataset containing prospective customers with combined demographic and address data.

Steps Performed

1. Data Loading & Initial Inspection

Loaded 1,000 records with 23 columns. Identified 5 irrelevant Unnamed: 16 through Unnamed: 20 columns.

2. Dropping Irrelevant Columns

Removed Unnamed: 16-20, Rank, and Value columns.

3. Missing Values Handling

Column	Missing Count	Missing %	Action Taken
last_name	29	2.9%	Filled with "None"
DOB	17	1.7%	Mode imputation
job_title	106	10.6%	Retained
job_industry_category	165	16.5%	Retained

4. Gender Standardization

Applied same standardization as Customer Demographics (Male, Female, U).

5. Feature Engineering

Created Age column from DOB. Created property_valuation_category by binning property valuation into Low/Medium/High.

6. Wealth Segment & Property Valuation Analysis

Analyzed distribution of wealth segments against property valuations. Cross-tabulated to find correlations.

7. Output

Final cleaned dataset: 1,000 records × 14 columns. Exported to NewCustomerList_Cleaned.csv.

Key Insights

- Wealth Segment Distribution: Majority are Mass Customer, with significant Affluent Customer and High Net Worth segments.
- Property Valuation Correlation: Higher wealth segments tend to correlate with higher property valuations.
- Geographic Distribution: Customers spread across NSW, VIC, and QLD.
- Car Ownership: Significant portion of customers own cars (useful for targeting in-store vs. online promotions).

Notebook 4: DQA and Data Cleaning — Transactions

Purpose: Cleans the Transactions dataset to ensure accurate sales and product data for downstream analysis.

Steps Performed

1. Data Loading & Initial Inspection

Loaded 20,000 records with 13 columns. Identified data type issue: product_first_sold_date stored as float (Excel serial date) instead of datetime.

2. Missing Values Analysis

Column	Missing Count	Missing %	Action Taken
online_order	360	1.8%	Mode imputation (filled with 1)
brand, product_line, product_class, product_size, standard_cost, product_first_sold_date	197 each	~1.0%	Dropped records (all null together)

3. Key Observation

When brand is null, all related product columns are also null. These 197 records (~1%) were dropped as they provide no actionable information.

4. Feature Engineering – Profit Calculation

Created new Profit column: Profit = list_price - standard_cost. Visualized profit distribution using histogram.

5. Data Type Corrections

Converted product_first_sold_date from float to datetime format.

6. Output

Final cleaned dataset: 19,803 records × 14 columns (including Profit). Exported to Transactions_Cleaned.csv.

Key Insights

- Online vs. Offline: Majority of transactions are online orders (online_order = 1).
- Order Status: All transactions in the dataset have "Approved" status.
- Profit Distribution: Right-skewed distribution — most products have moderate profit margins with some high-profit outliers.
- Brand Distribution: Top brands include Solex, Trek Bicycles, Giant Bicycles, Norco Bicycles, and OHM Cycles.

Notebook 5: RFM Analysis

Purpose: Performs RFM (Recency, Frequency, Monetary) analysis to segment customers based on their purchasing behavior for targeted marketing and business strategy.

What is RFM Analysis?

RFM is a behavior-based customer segmentation methodology:

Metric	Definition	Business Meaning
Recency (R)	Days since last purchase	How recently did the customer buy?
Frequency (F)	Total number of purchases	How often do they buy?
Monetary (M)	Total money spent (profit)	How much do they spend?

Steps Performed

1. Data Merging

Merged Transactions_Cleaned.csv with CustomerDemographic_Cleaned.csv on customer_id. Result: 19,354 records × 26 columns.

2. Date Conversion

Converted transaction_date to datetime format.

3. RFM Table Creation

Recency: Days between each customer's last purchase and the maximum transaction date (2017-12-30).

Frequency: Count of purchases per customer. Monetary: Sum of profit per customer.

```
rfm_table = merged_trans_cust.groupby(['customer_id']).agg({
    'transaction_date': lambda date: (comparison_date - date.max()).days,
    'product_id': lambda prod_id: len(prod_id),
    'Profit': lambda p: sum(p)
})
```

4. Quartile Scoring

Divided R, F, M into 4 quartiles (scores 1-4). Recency: Lower days = higher score (4 = most recent). Frequency: Higher count = higher score (4 = most frequent). Monetary: Higher spend = higher score (4 = highest spender).

5. RFM Score Calculation

Combined scores: RFM Score = $100 \times R + 10 \times F + 1 \times M$. Maximum possible score: 444 (best customers). Minimum possible score: 111 (worst customers).

6. Customer Segmentation

Customers divided into 4 tiers based on RFM score:

Tier	Label	Description
4 (Highest)	Platinum	Highest RFM scores — most valuable customers
3	Gold	Mid-high RFM scores — strong performers

2	Silver	Mid-low RFM scores — moderate engagement
1 (Lowest)	Bronze	Lowest RFM scores — least valuable

Key Insights & Business Value

Strategic Actions by Segment

Segment	Characteristics	Recommended Action
Platinum	Bought recently, frequently, high amounts	Loyalty rewards, exclusive offers, early access
Gold	Good on 2 of 3 metrics	Personalized recommendations, bundle deals
Silver	Moderate activity	Email campaigns, discounts to increase frequency
Bronze	Infrequent, low spend, long ago	Win-back campaigns, surveys to understand disengagement

Business Questions Answered

Q: Who are my best customers?

A: Platinum & Gold segments

Q: Who is at risk of churning?

A: Bronze segment (low recency)

Q: Who should receive upsell offers?

A: Silver & Gold (high recency, moderate frequency/monetary)

Q: Where should marketing budget be focused?

A: Retain Platinum, activate Silver

Data-Driven Insights

- High Recency + High Frequency = Loyal Advocates — These customers are brand ambassadors.
- High Monetary + Low Frequency = Big Spenders — Occasional but valuable; increase engagement.
- Low Recency + High Monetary = At-Risk Valuable Customers — Immediate win-back needed.

Tools & Technologies

Technology	Purpose
Python 3.x	Programming language
Pandas	Data manipulation, cleaning, and analysis
NumPy	Numerical operations
Matplotlib & Seaborn	Data visualization (distributions, histograms)
Jupyter Notebook	Interactive development environment
openpyxl	Excel file reading

Summary of Data Cleaning Impact

Dataset	Before	After	Records Changed
Customer Address	3,999	3,999	0 (state standardization only)
Customer Demographics	4,000	3,912	-88 (DOB null records dropped)
New Customer List	1,000	1,000	0 (imputation applied)
Transactions	20,000	19,803	-197 (null brand records dropped)

Key Takeaways

- 1. End-to-End Data Pipeline:** Demonstrates complete data quality assessment → cleaning → analysis workflow.
- 2. Data Quality Skills:** Missing value handling, inconsistency detection, outlier management, data type conversion.
- 3. Feature Engineering:** Created Age, Profit, property_valuation_category, and RFM scores.
- 4. Business Acumen:** RFM analysis translates directly to actionable marketing strategies.
- 5. Statistical Understanding:** Quartile-based segmentation, distribution analysis, correlation detection.
- 6. Technical Proficiency:** Python, Pandas, visualization libraries, Jupyter ecosystem.

Potential Future Work

- Predictive Modeling: Build churn prediction models using RFM scores as features.
- Customer Lifetime Value (CLV): Estimate long-term customer value.
- A/B Testing: Test targeted marketing campaigns on different RFM segments.
- Geospatial Analysis: Map customer distribution and property valuations.
- Time-Series Analysis: Track RFM score evolution over time.
- Machine Learning Clustering: Apply K-Means or DBSCAN for unsupervised segmentation.

Data Analytics Project — Customer Segmentation & RFM Analysis
For educational and portfolio purposes.